

Problem

In long-context LLMs, evicting parts of the KV-cache can either **HELP** or **HURT**

It HELPS (dilute-prone tasks):

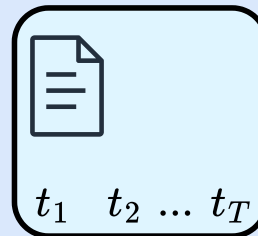
- Many redundant prefill tokens dilute attention, removing them improves accuracy
- e.g., summarization, QA, code, etc

It HURTS (capacity-bound tasks):

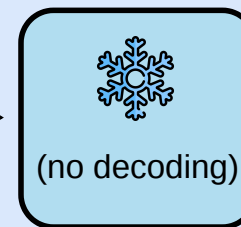
- Answer depend on precise retrieval of a few specific tokens
- Eviction causes significant failure

Setup: Prefill → Attention Analysis

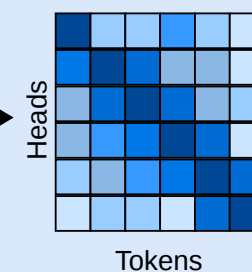
Long Input
(tokens)



LLM
(prefill-only)



Per-layer, Per-head
attention



Layers

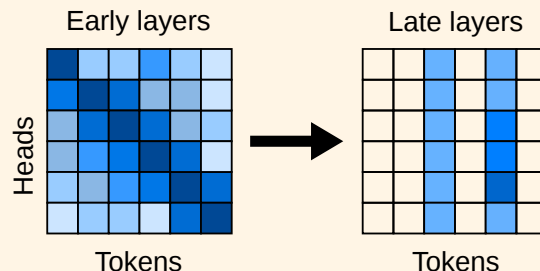
Compute D
from prefill
attention only

Intuition: Why D Works

Dilution-prone (large D)

Early layers: heads disagree (diffuse attention)

Late layers: heads agree on a few token

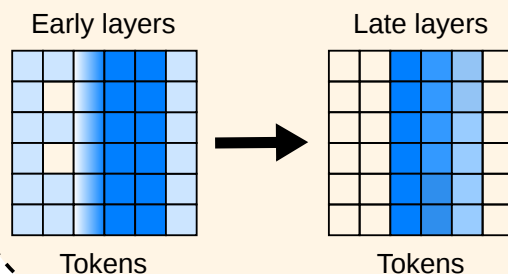


Eviction removes redundancy
→ attention concentrates on
important tokens → helps



Capacity-bound (small D)

Heads already agree early and stay agreed



Eviction likely drops few
critical tokens → accuracy
collapses



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